

Abstract

The Sound Reactive Light System is an analog electronics project designed to produce dynamic lighting effects that respond directly to audio input. By analyzing multiple frequency bands of background music, the system drives LEDs to create real-time visual patterns that enhance ambiance at events or parties. The design employs key analog building blocks—including band-pass filters, envelope detectors, and comparators—to isolate and process specific sound frequencies. Through this approach, the project demonstrates essential concepts in signal processing and circuit design while delivering an engaging and interactive audio-visual experience.

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1 Introduction and Functionality

The Sound Reactive Light System is an analog electronic project designed to create dynamic lighting effects that respond directly to audio signals. By analyzing different frequency bands of background music, the system drives a set of LEDs to change patterns in real time, enhancing the ambiance and visual appeal at parties or events. Using analog circuits such as low-pass, band-pass, high-pass filters, amplifiers, the design processes sound frequencies will cover the concepts in signal processing and circuit design while delivering an engaging interactive experience.

- System will capture sound and amplify in real time operation continuously.
- The audio signal is split into three main frequency bands: bass, midrange, and treble.
- All the three bands are amplified and used for LED driving.
- LEDs light up or change patterns based on the intensity and presence of sound in their respective frequency bands.

2 System Architecture

Functional Block Diagram

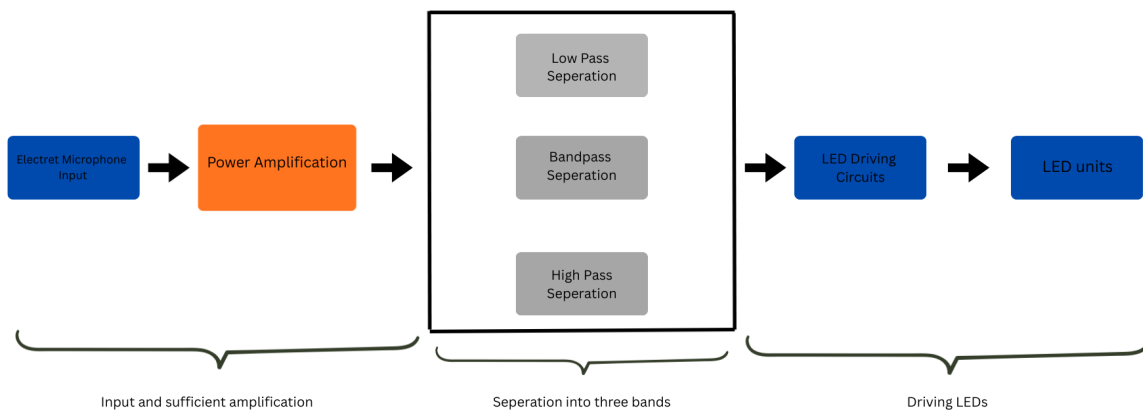


Figure 1: This is a full-width image.

1. Input

A **Electret Condensor Microphone** is used to take the music inputs into the system. And a DC blocking capacitor of value 1uF in series.

2. Power Amplification

A OpAmp amplification circuit is used to get the enough power amplification to the microphone signal in order to drive the rest of the circuit.

3. Three band separation

The input audio signal is separated into three bands of Lowpass, Bandpass and Highpass using Active OpAmp filters.

4. LED switching

Switching LEDs according to the output power of each filter.

3 Component Selection

3.1 Component justification

Component	Functionality	Reason
Filtering		
NE5532P OpAmp	Active filtering	Less distortion
LM358N OpAmp	Converts AC to DC	Less distortion
Power Supply		
7805 voltage regulator	To provide +5v +VCC to opamps and LEDs	accuracy of about $\pm 4\%$
ICL 7660S power inverter	Invert +5V to get -5V -VCC	Improves noise reduction
9V battery	Provide 9V DC power	availability and easy replacement
Input and Amplification		
Condensor microphone	Accepts music input signal	High sensitivity, compact size and low power requirements
NE5532P OpAmp	Amplifies input signal	Low noise of 15nV and less distortion than other tested LM324 and TL072

Table 1: Circuit components categorized by functional sections

3.2 Cutoff frequency calculations

1. **Low-Pass Filter** Cutoff frequency:

$$\begin{aligned}
 f &= \frac{1}{2\pi RC} \\
 &= \frac{1}{2\pi \cdot 270 \cdot 2.2 \cdot 10^{-6}} \\
 &= 267Hz
 \end{aligned}$$

2. **High-Pass Filter** Cutoff frequency:

$$\begin{aligned}
 f &= \frac{1}{2\pi RC} \\
 &= \frac{1}{2\pi \cdot 10 \cdot 10^3 \cdot 4.7 \cdot 10^{-9}} \\
 &= 3386Hz
 \end{aligned}$$

3. **Band-Pass Filter**

- (i) Low cutoff frequency:

$$\begin{aligned}
 f &= \frac{1}{2\pi RC} \\
 &= \frac{1}{2\pi \cdot 220 \cdot 2.2 \cdot 10^{-6}} \\
 &= 328Hz
 \end{aligned}$$

- (ii) High cutoff frequency:

$$\begin{aligned}
 f &= \frac{1}{2\pi RC} \\
 &= \frac{1}{2\pi \cdot 20 \cdot 10^3 \cdot 3.3 \cdot 10^{-9}} \\
 &= 2411Hz
 \end{aligned}$$

4 PCB Design

The printed circuit board (PCB) for the Sound Reactive Light System was designed using **Altium Designer**, which enabled accurate schematic capture and efficient layout development. Surface-mount device (SMD) components were selected to achieve a compact and space-efficient design while reducing parasitic effects and improving signal integrity. A **two-layer PCB** was implemented to simplify routing, provide proper grounding, and minimize noise in sensitive analog signal paths. After completing all design rule checks (DRC) and final verification, the PCB was fabricated and assembled through **JLCPCB**, ensuring reliable manufacturing quality and dimensional accuracy.

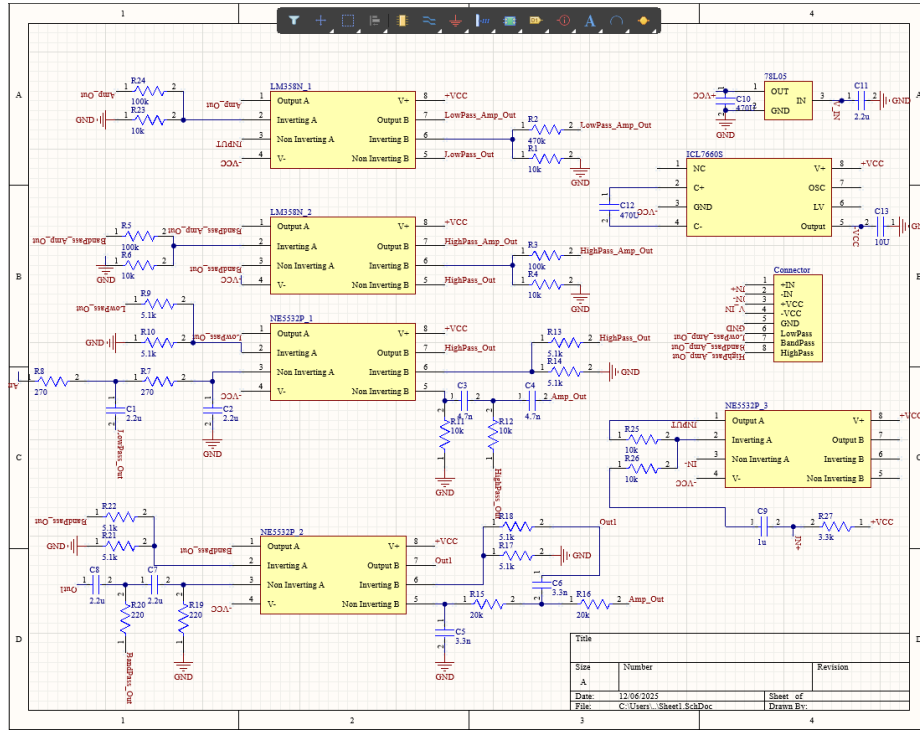
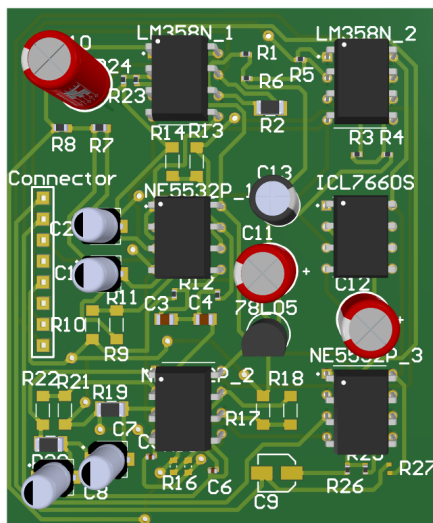
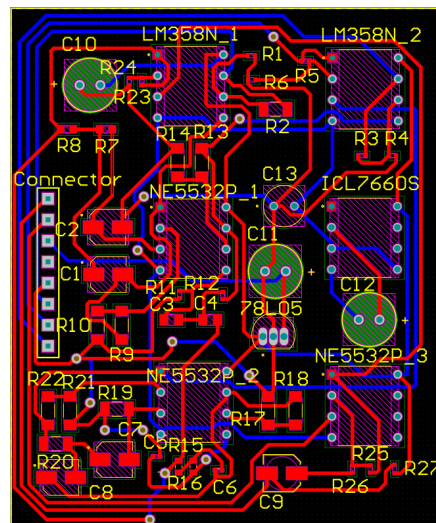


Figure 2: PCB Schematic



(a) 3D view PCB



(b) Routine of PCB

5 Enclosure Design

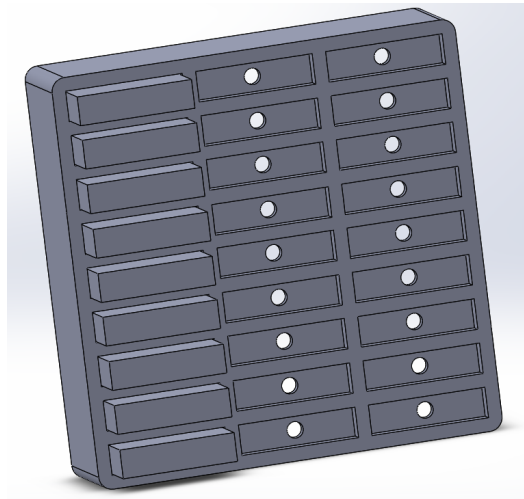
The enclosure for the Sound Reactive Light System was designed to provide a durable, safe, and visually appealing housing for the electronic circuitry and LED display. Its primary purpose is to protect internal components from physical damage, minimize electrical interference, and enhance the overall user experience by presenting a clean and organized final product.

The enclosure was constructed using PLA (polylactic acid), chosen for its strength, ease of fabrication, and cost-effectiveness. This material also offers good thermal stability and provides an aesthetically pleasing finish suitable for a display-oriented project.

The low thickness used in LED boxes allowed the LED effects to be visible while safeguarding the electronics inside.

Structural Layout

The internal layout was designed to ensure efficient use of space and proper separation of functional blocks. The PCB containing the band-pass filters, envelope detectors, and comparators was mounted securely using standoffs to prevent vibration and short circuits. The LED arrays were positioned along the front panel for maximum visibility, while wiring channels were created to keep signal lines organized and prevent noise coupling. A dedicated compartment was provided for the power supply to minimize interference with sensitive analog circuits.



(a) Designed Enclosure



(b) Enclosure of the product

6 Software Simulation and Hardware Testing

The frequency response of the Sound Reactive Light System was simulated using **LTspice**. Each of the filters—low-pass, band-pass, and high-pass—was modeled with the actual component values chosen for the design. The simulations allowed us to verify that the **cutoff frequencies** of the filters matched the theoretical calculations, ensuring that each frequency band would correctly trigger its corresponding LED channel. This process provided confidence that the system would perform as expected before hardware implementation.

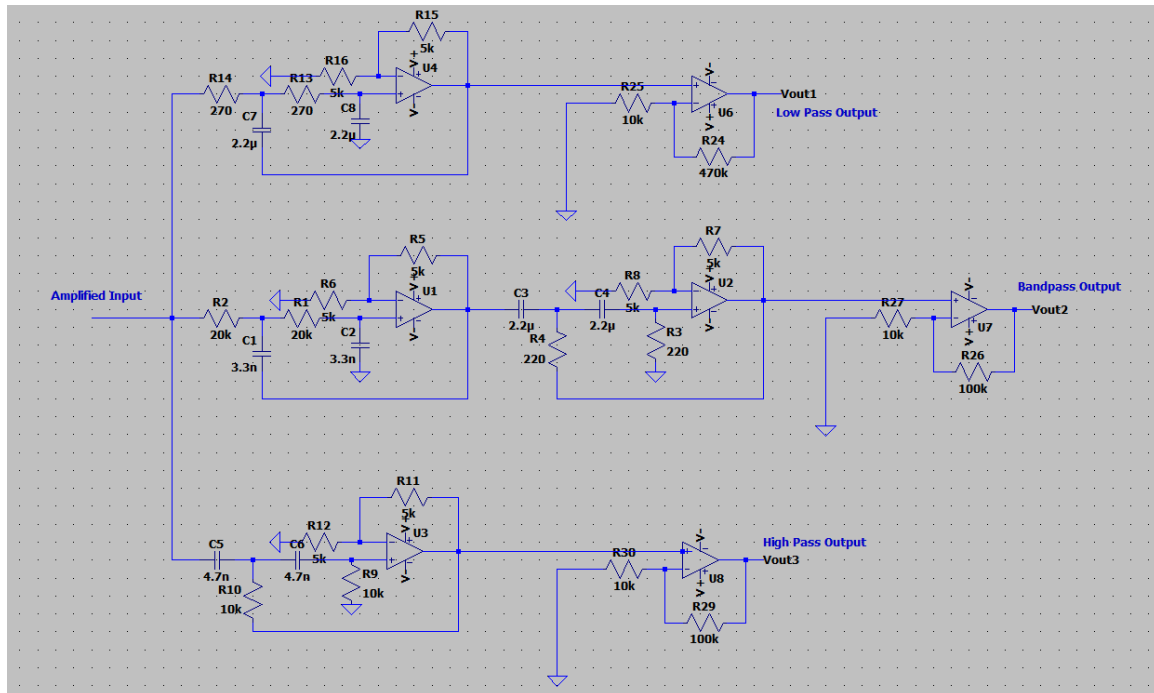


Figure 5: Circuit diagram showing the low-pass, band-pass, and high-pass filter implementation.

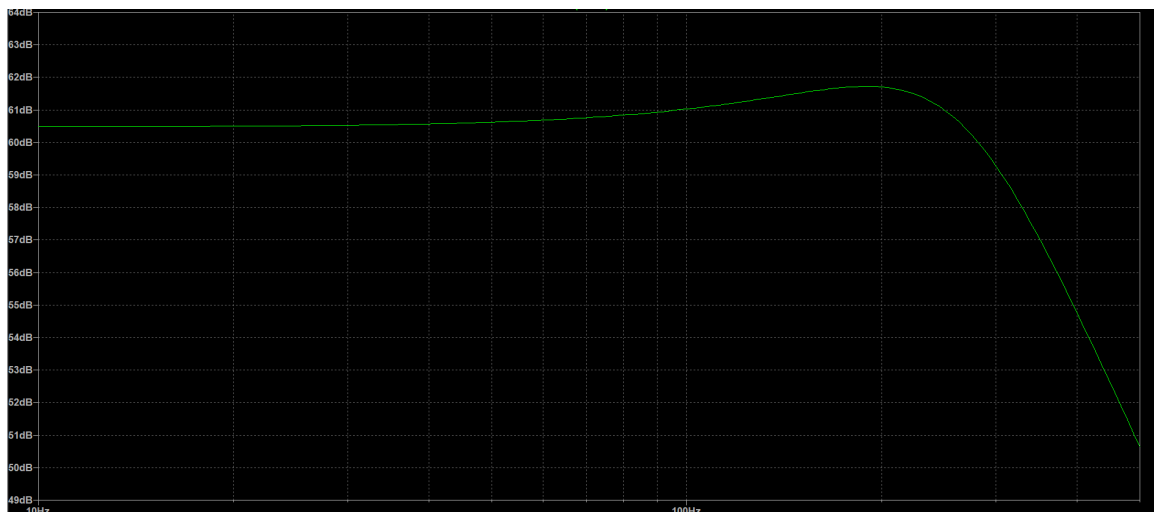
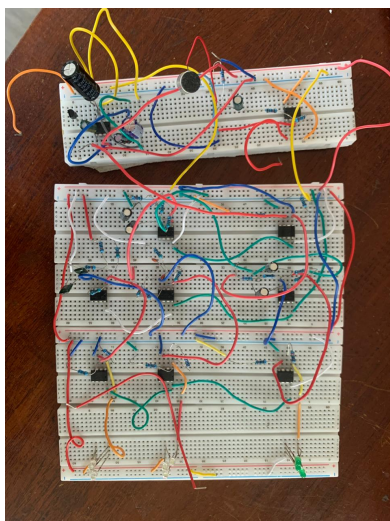
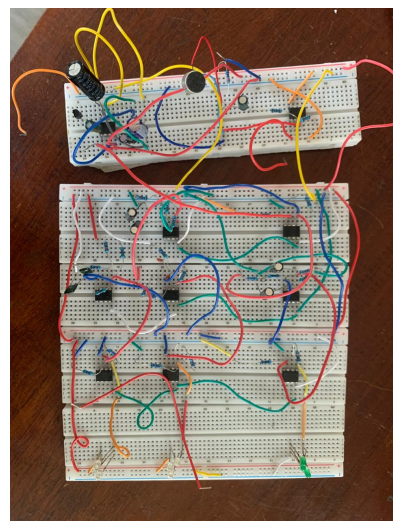


Figure 6: Simulation result: Low-pass filter frequency response.



(a) Breadboard implementation



(b) Breadboard implementation

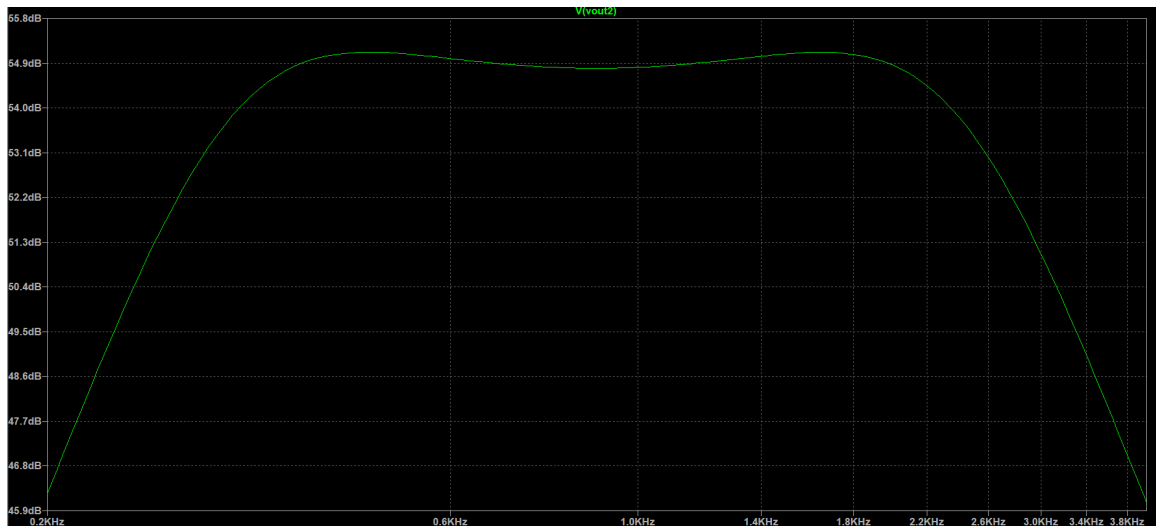


Figure 7: Simulation result: Band-pass filter frequency response.



Figure 8: Simulation result: High-pass filter frequency response.

7 Conclusion and Future Works

This project successfully demonstrated the design and implementation of an analog sound-reactive disco light system using only discrete and analog components. Audio signals captured by an electret condenser microphone were amplified, filtered, and processed to control the lighting circuit in response to sound intensity. Hardware testing confirmed stable and reliable operation with effective noise reduction. The compact two-layer PCB designed using Altium with SMD components satisfied all functional requirements and met the project objectives. **Future Developments**

1. Implementing a 3.5mm audio jack
 -Using a 3.5mm audio port for inputs where user can switch between either microphone input or audio port input. Which enables more flexibility in high noise environment.

2. Enclosure improvements - Using a adjustable stand which can adjust the angle of the LED arrays which give a better experience.

8 Contribution of Group Memembers

Index	Name	Tasks
230564L	Samarasekara S.M.R.P.	Software simulation and Soldering
230268H	Jayakody J.A.C.P.	Enclosure Design and PCB Design
230280L	Jayasinghe J.A.P.R	Breadboard implementation and PCB Design
230327N	Kaushalya R.G.S.P.	Breadboard implemntation and PCB Design

Table 2: Group members and their assigned tasks

9 References

1. The power supply for this project was designed according to the guidelines provided on <https://bestengineeringprojects.com/dual-polarity-5v-from-9v-battery/>.
2. OpAmp Datasheets <https://www.alldatasheet.com/datasheet-pdf/pdf/17968/PHILIPS/NE5532.html>,
<https://www.alldatasheet.com/datasheet-pdf/pdf/11678/ONSEMI/LM358N.html>
3. Voltage Regulators <https://www.alldatasheet.com/datasheet-pdf/pdf/69437/KEC/7805.html>,
<https://www.alldatasheet.com/datasheet-pdf/pdf/1046626/RENESAS/ICL7660.html>